Git + Linux Commands Reference (Linux/macOS vs Windows)

1. Git Commands

Purpose	Linux / macOS (bash, zsh, Git Bash)	Windows PowerShell
Initialize repo	`git init`	`git init`
Check status	`git status`	`git status`
Add all files	`git add .`	`git add .`
Add one file	`git add file1.txt`	`git add file1.txt`
Commit	`git commit -m "first commit"`	`git commit -m "first commit"`
View commit history	`git log`	`git log`
One-line log	`git log --oneline`	`git log --oneline`
Show branches	`git branch`	`git branch`
Create branch	`git branch feature`	`git branch feature`
Switch branch	`git checkout feature`	`git checkout feature`
New switch syntax	`git switch feature`	`git switch feature`
Create + switch branch	`git checkout -b feature`	`git checkout -b feature`
Merge branch	`git merge feature`	`git merge feature`
Delete branch	`git branch -d feature`	`git branch -d feature`
Clone repo	`git clone <url>`	`git clone <url>`
Connect remote	`git remote add origin <url>`	`git remote add origin <url>`
Push code	`git push origin main`	`git push origin main`
Pull code	`git pull origin main`	`git pull origin main`
Fetch updates	`git fetch`	`git fetch`
Show remotes	`git remote -v`	`git remote -v`

Purpose	Linux / macOS (bash, zsh, Git Bash)	Windows PowerShell
Restore file	`git restore file1.txt`	`git restore file1.txt`
Unstage file	`git restore --staged file1.txt`	`git restore --staged file1.txt`
Remove tracked file	`git rm file1.txt`	`git rm file1.txt`
Rename file	`git mv old.txt new.txt`	`git mv old.txt new.txt`
View differences	`git diff`	`git diff`
Configure username	`git config --global user.name "Name"`	same
Configure email	`git config --global user.email "mail@example.com"`	same
See config	`git config --list`	same

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2. Linux / Terminal Commands

Purpose	Linux / macOS / Git Bash	Windows PowerShell
Current folder	`pwd`	`pwd`
List files	`ls`	`ls`
Show hidden files	`ls -la`	`ls -Force`
Change folder	`cd foldername`	`cd foldername`
Go back	`cd ..`	`cd ..`
Go home	`cd ~`	`cd ~`
Create folder	`mkdir test`	`mkdir test`
Remove empty folder	`rmdir test`	`rmdir test`
Remove folder recursively	`rm -r test`	`Remove-Item test -Recurse`
Create file	`touch file1.txt`	`New-Item file1.txt -ItemType File`
Delete file	`rm file1.txt`	`Remove-Item file1.txt`
Copy file	`cp a.txt b.txt`	`Copy-Item a.txt b.txt`
Move file	`mv a.txt folder/`	`Move-Item a.txt folder/`

Purpose	Linux / macOS / Git Bash	Windows PowerShell
Rename file	`mv old.txt new.txt`	`Rename-Item old.txt new.txt`
Read file	`cat file1.txt`	`Get-Content file1.txt`
Clear terminal	`clear`	`cls`
Search text	`grep word file.txt`	`Select-String word file.txt`
Command history	`history`	`Get-History`
Open current folder	`code .`	`code .`

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3. Variables (Important Difference)

Purpose	Linux / bash	Windows PowerShell
Save output to variable	`sha1=$(git hash-object -w file.txt)`	`$sha1 = git hash-object -w file.txt`
Print variable	`echo $sha1`	`echo $sha1`
Commit-tree example	`commit=$(echo "Initial commit" | git commit-tree $tree)`	`$commit = echo "Initial commit" | git commit-tree $tree`

Best Recommendation

For learning Git from your tutor:

Use **Git Bash** on Windows.

Because most Linux/macOS commands will work exactly the same there.

For Windows system work:

Use **PowerShell**.

This avoids confusion and makes tutorials much easier to follow.